

September Exam session Projects List

Objectives

The goal of the project is to tackle one of the proposed topics in the field of Computer Vision, developing non-trivial solutions. Students should explore the problem from an original perspective, applying methods that go beyond conventional solutions and demonstrating critical thinking and problem-solving skills.

Deadline

Students willing to deliver their project in the September exam session must fill [this Google Form](#) by September 6th.

Lines of conduct

- **Student groups:** The project can be carried out by a group, each consisting of a maximum of 3 people. Projects can also be completed by individual students, but we suggest to work in team.
- **Notebook format:** The project must be implemented using a notebook (e.g., Google Colab, Kaggle) or with an IDE (e.g., VSCode, PyCharm). The code should be optimized to support GPU usage and run without any error. The delivered code must follow the structure outlined below:
 - *Imports:* all the needed packages (for the notebook format)
 - *Globals:* useful variables on the whole code
 - *Utils:* code support functions
 - *Data:* everything related to data management
 - *Network:* code to structure the neural network
 - *Train:* part containing the training cycle elements
 - *Evaluation:* tests needed for the trained network

You can find a sample template at this [link](#). Try to maintain as much as possible this conceptual structure.

- **Deep learning framework:** All projects **MUST** be done in Python via the Pytorch framework.
- **Project assignment:** You are required to choose a project through [this Google Form](#). In this form, you will provide information about your team and the project, and include the link to the project's GitHub repository. In this repository you have to upload:
 - Code (or notebook) implementing the project
 - Dataset (or a link to it)
 - Project presentation
 - Detailed README to provide a quick overview of the project and instructions on how to run it
- **Project submission:** The project must be presented on one of the exam dates. It can be presented at a different time than the written exam. Both the written exam and the project **MUST** be completed within the academic year (i.e., between the June 2026 session and the March 2027 session).
- **Plagiarism:** Any attempt to plagiarize, whether by copying other students' work, directly replicating code from online resources, or submitting content highly retrieved from generative AI models, will be strictly penalized. This course values originality and personal effort; therefore, students must submit independently developed solutions. On the other hand, it is acceptable to consult external resources for inspiration or guidance. Violations will be subject to academic disciplinary actions, potentially including a failing grade for the project or further measures outlined by university policy.

Project 6: Frequency-Domain Diffusion Models: Synthesizing Spectral Fingerprints for Deepfake Analysis

Abstract: Deepfakes and AI-generated images, despite achieving impressive photorealism in the spatial domain, consistently exhibit structural artifacts and tell-tale "fingerprints" in the frequency domain. These spectral anomalies are widely exploited by state-of-the-art deepfake detectors. This project explores a novel generative paradigm: training Diffusion Models directly on the frequency spectrum (e.g., via Discrete Cosine Transform or Fourier Transform) rather than the standard RGB spatial domain. By modeling the true spectral distribution of real images, the goal is to synthesize realistic spectra that, once inversely transformed, produce coherent RGB images. Students will investigate whether this approach can bridge the gap between generation and forensics by synthesizing images devoid of traditional generative frequency artifacts.

Dataset: FaceForensics++, Celeb-DF, GenImage, or standard high-quality facial datasets like FFHQ/CelebA. (*Or what you can run on your hardware*)

Task: The objective is to implement a pipeline that transforms a dataset of images into their frequency representations and trains a baseline Diffusion Model (DDPM) to generate these spectral maps from noise. Students must implement the forward and inverse frequency transforms, adapt the UNet architecture to handle complex or multi-channel frequency data (e.g., magnitude/phase or real/imaginary components), and reconstruct spatial RGB images from the generated spectra. Otherwise implement your own pipeline coherently with the idea of the proposed project. It is required to evaluate both the visual quality of the reconstructed images and their spectral properties compared to standard RGB-trained models.

Main objectives:

- *Frequency Transformation Pipeline:* Implement a robust preprocessing step to convert spatial images to the frequency domain (e.g., 2D-DCT or 2D-FFT) and format the data appropriately for neural network processing.
- *Anatomical Guidance Integration:* Formulate and integrate an algorithmic improvement (e.g., DDIM sampling strategy, a latent-space approach, progressive distillation, or something new!) aimed at accelerating inference or training.
- *Spectral Diffusion Training:* Train a baseline diffusion model entirely in the frequency domain. Address architectural challenges, such as balancing the high-variance low-frequency components with the sparse high-frequency details.
- *Reconstruction & Visual Fidelity:* Invert the generated spectra back to the spatial domain. Evaluate generation quality using standard quantitative metrics (FID, IS) and visually inspect the structural coherence of the resulting RGB images.
- *Forensic Spectral Analysis:* Compare the frequency spectra of the images generated by your model against those generated by a standard spatial DDPM. Conduct a comparative analysis to determine if the frequency-domain model successfully mitigates the typical grid-like artifacts or "synthetic fingerprints" common in spatial generative models.

Otherwise implement your own pipeline coherently with the idea of the proposed project.

References:

- 1 Frank, J., et al. "Leveraging Frequency Analysis for Deep Fake Image Recognition." ICML 2020.
- 2 Corvi, R., et al. "On the detection of synthetic images generated by diffusion models." ICASSP 2023.

- 3 Xu, L., et al. "Text-to-image with frequency domain diffusion models." SPIE 2025.
- 4 Song, T., et al. "FilterDiff: Noise-free Frequency-domain Diffusion Models for Accelerated MRI Reconstruction." MICCAI 2025.